

# Gabriel Agostini

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## EDUCATION

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| <b>Cornell University</b> , Cornell Tech<br><i>PhD in Information Science with minor in Urban Planning</i><br>Advisors: <a href="#">Emma Pierson</a> & <a href="#">Nikhil Garg</a> | 2022 - present |
| <b>Columbia University</b> , Columbia College<br><i>Bachelor of Arts in Urban Studies</i>   GPA: 4.06/4.00                                                                         | 2021 - 2022    |
| <b>Columbia University</b> , School of Engineering and Applied Sciences<br><i>Bachelor of Science in Applied Mathematics</i>   GPA: 4.17/4.00                                      | 2017 - 2021    |
| <b>Budapest Semesters in Mathematics</b>                                                                                                                                           | 2020           |

## FULL PAPERS

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**Gabriel Agostini**, Rachel Young, Maria Fitzpatrick, Nikhil Garg, Emma Pierson. "[Inferring fine-grained migration patterns across the United States](#)." *Nature Communications* 17 (1265), 2026.

**Gabriel Agostini**, Emma Pierson, Nikhil Garg. "[A Bayesian Spatial Model to Correct Under-Reporting in Urban Crowdsourcing](#)." *38th Annual AAAI Conference on Artificial Intelligence (AAAI24)*. **Oral presentation**. 2024.

Rajiv Movva\*, Sidhika Balachandar\*, Kenny Peng\*, **Gabriel Agostini\***, Nikhil Garg, Emma Pierson. "[Topics, Authors, and Institutions in Large Language Model Research: Trends from a Survey of 17K arXiv Papers](#)." *Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*. 2024

## CONFERENCE PAPERS

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Talia Kaufmann\*, **Gabriel Agostini\***, Trivik Verma, Dan O'Brien. "[Inequality in mobility patterns: reconciling access and travel in American cities](#)." *19th International Conference on Computational Urban Planning and Urban Management (CUPUM)*, London, United Kingdom. June 19, 2025.

**Gabriel Agostini**, Juliana Gonçalves, Trivik Verma. "[Identifying Urban Morphology from Street Networks with Graphlet Analysis](#)." *30th Annual Geographical Information Science Research UK (GISRUK)*, Liverpool, United Kingdom. April 1, 2022.

**Gabriel Agostini**, Yushan Zhang, Debra Laefer. "[A Category-Theory Approach to Construction Ontologies in Subsurface Mass Transit](#)." *International Archives of Photogrammetry, Remote Sensing, and Spatial Information Sciences*, XLVI-4/W4-2021, 125–130, 2021.

## IN PREPARATION

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**Gabriel Agostini**, Nikhil Garg, Emma Pierson. "Modeling Biker Preferences with Large Scale E-Bike GPS Data". To appear in *EAAMO 2026* (non-archival).

**Gabriel Agostini**, Zhixuan Qi, Katelyn Liu, Shreya Sriram, Esther Li, Emma Pierson, Nikhil Garg. "Quantifying Spatiotemporal Gaps in Accessibility with Fine-Grained, Open-Source Amenity Datasets"

Erica Chiang, Katelyn Liu, Anum Ahmad, Zhi Liu, **Gabriel Agostini**, Tyler Gibson, Uma Bhandaram, Nikhil Garg. "The Forest from the Trees: Lessons on Public-Sector Deployment from an Academic Collaboration with NYC Department of Parks and Recreation"

## PUBLIC SECTOR EXPERIENCE

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**New York City Department of Transportation Cycling & Micromobility** June 2025 - Present  
*Fellow, College Aide | Advisors: **Carl Sundstrom**, **Mark Seaman***

- Built an automated pipeline to simplify NYC street network and incorporate bike facilities
- Developed a general map-matching algorithm suitable for large-scale GPS ping data.

**Stanford & San Francisco Unified School District Partnership** June 2021 - December 2021  
*Stanford University Data Science for Social Good | Advisor: **Prof. Irene Lo***

- Designed algorithms to ensure equal access to education for underserved students in San Francisco.
- Employed spatial analysis techniques to identify geographical proxies for educational barriers.

## TEACHING EXPERIENCE

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**Cornell Tech**, Teaching Assistant Fall 2022, Fall 2023, Spring 2024  
*Courses: Applied Data Science, Fairness in Machine Learning Algorithms*

**Columbia University Mathematics Department**, Teaching Assistant Spring 2019 - Spring 2022  
*Courses: Multivariable Calculus, Linear Algebra, Ordinary Differential Equations*

**Fênix Pré-Vestibulares**, Mathematics Instructor Summer 2018

## COLLABORATIONS IN THE ARTS

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**Backslash at Cornell Tech**, Associate producer for BADWIZARD September 2025 - Present  
*Artist collaborator: Tianyi Sun*

**PiTech Arts and Culture Fellowship** September 2025 - May 2026  
*Artist collaborators: Lincoln Center Collider Fellows*

## AWARDS & HONORS

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**IPUMS Spatial Student Research Award**, IPUMS April 2026

**International Geographic Information Funds Student Paper Award**, AAG-IGIF March 2026

**Phi Beta Kappa**, Columbia College April 2022

**Summa cum Laude**, Columbia College April 2022

**Summa cum Laude**, Columbia University School of Engineering and Applied Sciences May 2021

## SERVICE & LEADERSHIP

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**Urban Data EAAMO-Bridges Working Group**, Organizer 2024 - Present  
*Equity and Access in Algorithms, Mechanisms, and Optimization (EAAMO)*

**EAAMO Conference**, Proceedings Chair 2025  
*Equity and Access in Algorithms, Mechanisms, and Optimization (EAAMO)*

**Cornell Information Science Graduate Application Review Program**, Mentor 2023, 2024, 2025  
*Cornell Information Science Student Council*

## INVITED TALKS

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|----------------------------------------------------------------------------------------|------------------------------|
| <b>Penn State University</b> , <a href="#">EEPI Seminar</a>                            | April 2026                   |
| <b>New York University</b> , Urban Lab: Immigration and Housing in Queens              | March 2026                   |
| <b>Vision Zero Initiative</b> , NYC Research on the Road                               | November 2025                |
| <b>Cornell Tech</b> , Urban Systems                                                    | September 2025               |
| <b>Barnard College Center for Computational Science</b> , <a href="#">CSC Workshop</a> | September 2025               |
| <b>Cornell Tech</b> , Applied Data Science                                             | October 2024, September 2025 |
| <b>Cornell Tech</b> , Urban Data Science                                               | November 2024                |

## FELLOWSHIPS

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|-----------------------------------------------------------------------------------------------------------------------|-------------------------|
| <b>PiTech Arts and Culture PhD Fellowship</b><br><i>PiTech: Public Interest Technology Initiative at Cornell Tech</i> | Fall 2025 - Spring 2026 |
| <b>Siegel-PiTech PhD Impact Fellowship</b><br><i>PiTech: Public Interest Technology Initiative at Cornell Tech</i>    | Summer 2025             |
| <b>Stanford Data Science for Social Good Fellowship</b><br><i>Stanford Data Science Institute</i>                     | Summer 2021             |
| <b>Tandon Undergraduate Summer Research Program</b><br><i>New York University Tandon School of Engineering</i>        | Summer 2020             |

## RELEVANT COURSEWORK

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|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Analysis &amp; Topology</i> | Real Analysis (A), Measure Theory (A+), Complex Analysis (A+), Functional Analysis (A)<br>ODEs (A+), PDEs (A+), Dynamical Systems (A+), Topology (A+)            |
| <i>Algebra</i>                 | Linear Algebra (A+), Abstract Algebra: Group Theory (A), Abstract Algebra II (A+)                                                                                |
| <i>Statistics</i>              | Probability Theory (A), Stochastic Processes (A+), Time Series Analysis (A+)<br>Statistical Inference (A+), Multivariate Statistics (A), Sample Surveys (A)      |
| <i>Network Science</i>         | Graph Theory (A+), Spectral Theory of Graphs and Groups (A+),<br>Structure of Complex Networks (A+), Analysis of Networks and Crowds (A)                         |
| <i>Machine Learning</i>        | Machine Learning (A), Geometric Data Analysis (A), Bayesian Optimization (A+)                                                                                    |
| <i>Urban Studies</i>           | Global Urbanism (A), Urban Elsewheres (A), History of the City in Latin America (A)<br>Colonial Cities of the Americas (A), Postwar Fiction and Architecture (A) |

## SKILLS

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|--------------------|---------------------------------------------------------------------------------------|
| <i>Programming</i> | Python (matplotlib, seaborn, scipy, pysal, networkx, geopandas, osmnx, sklearn), Stan |
| <i>Technology</i>  | Git, Microsoft Excel, ArcGIS, QGIS                                                    |
| <i>Languages</i>   | Brazilian Portuguese (native), Italian (advanced), Russian (basic)                    |